Numerical Analysis

Chapter 1: Mathematical Review

Department of Mathematics & Statistics
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Textbook: Classical and Modern Numerical Analysis:
Theory, Methods and Practice
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CRC Press, Boca Raton, FL, 2010

Outline

1. Notations
2. Review for calculus
3. Review for linear algebra

Outline

1 Notations

2 Review for calculus

3 Review for linear algebra

Notations

- Let $C[a, b]$ denote the function space of continuous functions on $[a, b]$.
- Let $C^k[a, b]$ denote the function space of kth differentiable functions on $[a, b]$. That is, if $f(x) \in C^k[a, b]$, then $f^{(i)}(x)$ $(i = 0, 1, \cdots, k)$ are continuous on $[a, b]$.
- Let $\mathbb{R}$ denote the space of real numbers. Let $\mathbb{C}$ denote the space of complex numbers.
- Let $\mathbb{R}^n$ denote the $n-$dimensional space of $n-$tuples of real numbers. Let $\mathbb{C}^n$ denote the $n-$dimensional space of $n-$tuples of complex numbers.
- Let P^k denote the space of the kth order polynomials. Let $P^k[a, b]$ denote the space of the kth order polynomials on $[a, b]$.

Outline

1 Notations

2 Review for calculus

3 Review for linear algebra

Intermediate value theorem

Theorem (I)

If f is a continuous function on $[a, b]$ and k is a number between $m = \min_{a \le x \le b} f(x)$ and $M = \max_{a \le x \le b} f(x)$, then there exists a number $c \in [a, b]$ such that $f(c) = k$.

Mean value theorem

Theorem (II)

Suppose $f \in C^1[a,b]$, $x \in [a,b]$, *and* $y \in [a,b]$. *Then there exists a* $\xi \in [x,y] \subseteq [a,b]$ *such that*

$$f(y) - f(x) = f'(\xi)(y-x).$$

Taylor's expansion

Theorem (III)

Suppose that $f \in C^{n+1}[a,b]$ *and* $x_0 \in [a,b]$. *Then for any* $x \in [a,b]$, *we have the following Taylor's expansion of* $f(x)$ *at* x_0:

$$f(x) = P_n(x) + R_n(x),$$

where

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k \\ &= f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \cdots \\ &\quad + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n, \end{aligned}$$

Taylor's expansion

Theorem (III: Continued)

$$R_n = \frac{1}{(n+1)!} f^{(n+1)}(\xi)(x-x_0)^{n+1}$$

for some $\xi \in [x_0, x]$ *(Lagrange form of the remainder)* ,

or

$$R_n = \frac{1}{n!} \int_{x_0}^{x} f^{(n+1)}(s)(x-s)^n \, ds \quad \textit{(Integral form of the remainder)} .$$

Cauchy sequence

Definition (Cauchy sequence)

A sequence $\{x_k\}_{k=1}^{\infty}$ is called a Cauchy sequence if, given any $\varepsilon > 0$, there exists a N dependent on ε such that $|x_k - x_n| < \varepsilon$ for every k and n greater than N.

Lemma (I)

Cauchy sequences in $\mathbb{R}$ and $\mathbb{C}$ must converge to a point in $\mathbb{R}$ or $\mathbb{C}$.

Definition (Complete spaces)

Number systems in which Cauchy sequences must converge are called complete spaces.

Outline

1 Notations

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3 Review for linear algebra

Matrix

Definition (Matrix)

An $m \times n$ matrix A is a rectangular array of elements arranged in m rows and n columns:

$$A = [a_{ij}]_{i=1,j=1}^{m,n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Algebraic operations on matrices

Definition (Transpose)

The transpose of the $m \times n$ matrix $A = [a_{ij}]_{i=1,j=1}^{m,n}$ is a $n \times m$ matrix obtained from A by interchanging the rows and columns of A:

$$A^T = [a_{ji}]_{j=1,i=1}^{n,m} = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \cdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

Algebraic operations on matrices

Definition (Conjugate)

The conjugate of a matrix A is a matrix obtained from A by replacing each element a_{ij} by its conjugate $\overline{a}_{ij}$:

$$\overline{A} = [\overline{a}_{ij}]_{i=1,j=1}^{m,n} = \begin{pmatrix} \overline{a}_{11} & \overline{a}_{12} & \cdots & \overline{a}_{1n} \\ \overline{a}_{21} & \overline{a}_{22} & \cdots & \overline{a}_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ \overline{a}_{m1} & \overline{a}_{m2} & \cdots & \overline{a}_{mn} \end{pmatrix}.$$

Definition (Adjoint)

The adjoint (conjugate transpose) of a matrix A is $A^H = \overline{A}^T$.

Algebraic operations on matrices

- Example 1: Compute the transpose, conjugate, and adjoint of

$$A = \begin{pmatrix} 3 & 2-i \\ 4+3i & -5+2i \end{pmatrix}.$$

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$$\overline{A} = \begin{pmatrix} 3 & 2+i \\ 4-3i & -5-2i \end{pmatrix}.$$

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- Adjoint:

$$A^H = \begin{pmatrix} 3 & 4-3i \\ 2+i & -5-2i \end{pmatrix}.$$

Algebraic operations on matrices

Definition (Equality)

Two $m \times n$ matrices $A = [a_{ij}]_{i=1,j=1}^{m,n}$ and $B = [b_{ij}]_{i=1,j=1}^{m,n}$ are said to be equal if all corresponding elements are equal. That is, $a_{ij} = b_{ij}$ for each i and j.

Definition (Zero)

A matrix is said to be 0 if its elements are all 0.

Definition (Multiplication by a number)

The product of an $m \times n$ matrix $A = [a_{ij}]_{i=1,j=1}^{m,n}$ by a number α is defined as $\alpha A = [\alpha a_{ij}]_{i=1,j=1}^{m,n}$.

Algebraic operations on matrices

Definition (Addition/Subtraction)

The sum/difference of two $m \times n$ matrices $A = [a_{ij}]_{i=1,j=1}^{m,n}$ and $B = [b_{ij}]_{i=1,j=1}^{m,n}$ are defined as $A \pm B = [a_{ij} \pm b_{ij}]_{i=1,j=1}^{m,n}$.

Definition (Matrix multiplication)

The product of an $m \times n$ matrix $A = [a_{ij}]_{i=1,j=1}^{m,n}$ and an $n \times r$ matrix $B = [b_{ij}]_{i=1,j=1}^{n,r}$ is an $m \times r$ matrix $C = [c_{ij}]_{i=1,j=1}^{m,r}$ where

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Algebraic operations on matrices

Properties of matrices:

- Commutative law for addition: $A + B = B + A$.
- Associative law for addition: $A + (B + C) = (A + B) + C$.
- Distributive law for multiplication by a number:

$$\alpha(A + B) = \alpha A + \alpha B,\ (\alpha + \beta)A = \alpha A + \beta A.$$

- Associative law for matrix multiplication: $(AB)C = A(BC)$.
- Distributive law for matrix multiplication: $A(B + C) = AB + AC$.
- In general, $AB \neq BA$.
- $(AB)^T = B^T A^T$, $(AB)^H = B^H A^H$.

Algebraic operations on matrices

- Example 2: Compute the sum and multiplication of the matrices

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 1 \end{pmatrix}, \; B = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}.$$

Algebraic operations on matrices

- Example 2: Compute the sum and multiplication of the matrices

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 1 \end{pmatrix}, \; B = \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}.$$

- Sum:

$$A + B = \begin{pmatrix} 3 & -1 & 0 \\ 1 & 1 & -1 \\ 4 & 0 & 2 \end{pmatrix}.$$

Algebraic operations on matrices

- Multiplication:

$$
\begin{aligned}
AB &= \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -1 \\ 2 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ 1 & -1 & 0 \\ 2 & -1 & 1 \end{pmatrix}. \\
&= \begin{pmatrix} 2-2+2 & 1+2-1 & -1+0+1 \\ 0+2-2 & 0-2+1 & 0+0-1 \\ 4+1+2 & 2-1-1 & -2+0+1 \end{pmatrix}. \\
&= \begin{pmatrix} 2 & 2 & 0 \\ 0 & -1 & -1 \\ 7 & 0 & -1 \end{pmatrix}.
\end{aligned}
$$

- $BA = ?$ Try it by yourself!

Determinant

Definition (Minor)

The minor associated with each element a_{ij} of a square matrix A is the determinant of the matrix obtained by deleting the ith row and the jth column of A.

Definition (Cofactor)

The cofactor associated with each element a_{ij} of a square matrix A is $(-1)^{i+j} M_{ij}$.

Determinant

Definition (Determinant)

The determinant of a $n \times n$ matrix A is defined as

$$det(A) = \sum_{i=1}^{n} a_{ij}(-1)^{i+j} M_{ij} \quad \text{for a fixed } j$$

or

$$det(A) = \sum_{j=1}^{n} a_{ij}(-1)^{i+j} M_{ij} \quad \text{for a fixed } i.$$

Remark

$$det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

Determinant

Remark

$$det(AB) = det(A)det(B).$$

Remark

For a lower triangular matrix

$$A = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ a_{n1} & a_{n2} & 0 & a_{nn} \end{pmatrix},$$

we also have $det(A) = \prod_{i=1}^{n} a_{ii}$.

Determinant

Remark

Similarly, for an upper triangular matrix

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix},$$

we have $det(A) = \prod_{i=1}^{n} a_{ii}$.

Determinant

- Example 3: Compute the determinant of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 1 & 8 \\ 1 & -2 & -7 \end{pmatrix}.$$

Determinant

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$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 1 & 8 \\ 1 & -2 & -7 \end{pmatrix}.$$

- By the definition,

$$det(A) = 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 8 \\ -2 & -7 \end{vmatrix} + 2 \cdot (-1)^{1+2} \begin{vmatrix} -2 & 8 \\ 1 & -7 \end{vmatrix}$$
$$+1 \cdot (-1)^{1+3} \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}$$

Determinant

- Example 3: Compute the determinant of

$$A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 1 & 8 \\ 1 & -2 & -7 \end{pmatrix}.$$

- By the definition,

$$\begin{aligned} det(A) &= 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 8 \\ -2 & -7 \end{vmatrix} + 2 \cdot (-1)^{1+2} \begin{vmatrix} -2 & 8 \\ 1 & -7 \end{vmatrix} \\ &\quad +1 \cdot (-1)^{1+3} \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} \\ &= (-7+16) - 2(14-8) + (4-1) \\ &= 9 - 12 + 3 \\ &= 0. \end{aligned}$$

Inverse matrix

Definition (Identity matrix)

The identity matrix I is a matrix whose elements are 1 on the diagonal and 0 everywhere else:

$$I = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Remark

For any square matrix A,

$$AI = IA = A.$$

Inverse matrix

Definition (Inverse)

A square matrix A is said to be nonsingular or invertible if there is another square matrix B such that $AB = BA = I$. And the matrix B is called the inverse of A and denoted by $B = A^{-1}$.

Remark

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Inverse matrix

Remark

- $(AB)^{-1} = B^{-1}A^{-1}$.
- $(A+B)^{-1} \neq A^{-1} + B^{-1}$.

Remark

The inverse matrix of an upper triangular matrix is an upper triangular matrix. Similarly, the inverse matrix of a lower triangle matrix is a lower triangular matrix.

Inverse matrix

- There are several different ways to compute the inverse matrix: row reduction, determinants,.....
- Row reduction: $(A, I) \Rightarrow (I, A^{-1})$ by using elementary row operations.
- Elementary row operations:

  (1)Interchanging of two rows.

  (2)Multiplication of a row by a nonzero scalar.

  (3)Addition of any multiple of one row to another row.
- We will use R_i to represent the ith row.

Inverse matrix

- Example 4: Find the inverse of

$$A = \begin{pmatrix} 1 & -1 & -1 \\ 3 & -1 & 2 \\ 2 & 2 & 3 \end{pmatrix}.$$

Inverse matrix

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$$A = \begin{pmatrix} 1 & -1 & -1 \\ 3 & -1 & 2 \\ 2 & 2 & 3 \end{pmatrix}.$$

- First,

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right).$$

Inverse matrix

- Example 4: Find the inverse of

$$A = \begin{pmatrix} 1 & -1 & -1 \\ 3 & -1 & 2 \\ 2 & 2 & 3 \end{pmatrix}.$$

- First,

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 3 & -1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right).$$

- Adding $-3R_1$ to R_2 and adding $-2R_1$ to R_3:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & -3 & 1 & 0 \\ 0 & 4 & 5 & -2 & 0 & 1 \end{array}\right).$$

Inverse matrix

- Adding $-2R_2$ to R_3:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & -3 & 1 & 0 \\ 0 & 0 & -5 & 4 & -2 & 1 \end{array}\right).$$

Inverse matrix

- Adding $-2R_2$ to R_3:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 5 & -3 & 1 & 0 \\ 0 & 0 & -5 & 4 & -2 & 1 \end{array}\right).$$

- Multiply R_2 by $\frac{1}{2}$ and multiply R_3 by $-\frac{1}{5}$:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Inverse matrix

- Adding $-\frac{5}{2}R_3$ to R_2 and adding R_3 to R_1:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 0 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Inverse matrix

- Adding $-\frac{5}{2}R_3$ to R_2 and adding R_3 to R_1:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 0 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

- Adding R_2 to R_1:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{7}{10} & -\frac{1}{10} & \frac{3}{10} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Inverse matrix

- Adding $-\frac{5}{2}R_3$ to R_2 and adding R_3 to R_1:

$$\left(\begin{array}{ccc|ccc} 1 & -1 & 0 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

- Adding R_2 to R_1:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{7}{10} & -\frac{1}{10} & \frac{3}{10} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

- Hence

$$A^{-1} = \left(\begin{array}{ccc} \frac{7}{10} & -\frac{1}{10} & \frac{3}{10} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{4}{5} & \frac{2}{5} & -\frac{1}{5} \end{array}\right).$$

Vector

Remark

A row vector $\overrightarrow{x} = (x_1, \cdots, x_n)$ *is a* $1 \times n$ *matrix. Similarly, a column vector*

$$\overrightarrow{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (x_1, \cdots, x_n)^T$$

is a $n \times 1$ *matrix.*

Remark

Let $\mathbb{R}^n$ *denote the* $n-$*dimensional space of* $n-$*tuples of real numbers. Let* $\mathbb{C}^n$ *denote the* $n-$*dimensional space of* $n-$*tuples of complex numbers. That is, if* $\overrightarrow{x} \in \mathbb{C}^n$*, then* $\overrightarrow{x} = (x_1, \cdots, x_n)^T$ *where* $x_i \in \mathbb{C}$ *for* $i = 1, \cdots, n$*.*

Linearly independence/dependence

Definition (Linearly independence/dependence)

The vectors $\overrightarrow{x}^{(1)}, \overrightarrow{x}^{(2)}, \cdots, \overrightarrow{x}^{(n)}$ are said to be linear independent if

$$\sum_{i=1}^{n} k_i \overrightarrow{x}^{(i)} = 0$$

implies $k_i = 0$ $(i = 1, 2, \cdots, n)$. Otherwise, $\overrightarrow{x}^{(i)}$ $(i = 1, 2, \cdots, n)$ are said to be linearly dependent.

Linearly independence/dependence

- Example 5: Determine whether the vectors

$$\vec{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \ \vec{x}^{(2)} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

are linearly independent or linearly dependent.

Linearly independence/dependence

- Example 5: Determine whether the vectors

$$\vec{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \ \vec{x}^{(2)} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

  are linearly independent or linearly dependent.

- Consider

$$k_1 \vec{x}^{(1)} + k_2 \vec{x}^{(2)} = 0.$$

Linearly independence/dependence

- Example 5: Determine whether the vectors

$$\vec{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \ \vec{x}^{(2)} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$$

  are linearly independent or linearly dependent.

- Consider

$$k_1 \vec{x}^{(1)} + k_2 \vec{x}^{(2)} = 0.$$

- Then

$$\begin{aligned} k_1 - 4k_2 &= 0, \\ 2k_1 + k_2 &= 0. \end{aligned}$$

Linearly independence/dependence

- Hence

$$2k_1 - 8k_2 = 0,$$
$$2k_1 + k_2 = 0.$$

Linearly independence/dependence

- Hence

$$2k_1 - 8k_2 = 0,$$
$$2k_1 + k_2 = 0.$$

- Then $-9k_2 = 0 \Rightarrow k_2 = 0 \Rightarrow k_1 = 0$.

Linearly independence/dependence

- Hence

$$2k_1 - 8k_2 = 0,$$
$$2k_1 + k_2 = 0.$$

- Then $-9k_2 = 0 \Rightarrow k_2 = 0 \Rightarrow k_1 = 0$.
- Therefore, $\overrightarrow{x}^{(1)}$ and $\overrightarrow{x}^{(2)}$ are linearly independent.

Linearly independence/dependence

- Example 6: Determine whether the vectors

$$\vec{x}^{(1)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \ \vec{x}^{(2)} = \begin{pmatrix} -4 \\ -8 \end{pmatrix}$$

  are linearly independent or linearly dependent.

- Consider

$$k_1 \vec{x}^{(1)} + k_2 \vec{x}^{(2)} = 0.$$

- Then

$$\begin{aligned} k_1 - 4k_2 &= 0, \\ 2k_1 - 8k_2 &= 0. \end{aligned}$$

Linearly independence/dependence

- Hence

$$2k_1 - 8k_2 = 0,$$
$$2k_1 - 8k_2 = 0.$$

- There is only one equation $2k_1 - 8k_2 = 0$ for two unknowns. Hence there are infinite many solutions.
- For example, pick $k_2 = 1$, then $k_1 = 4$.
- Therefore, $\overrightarrow{x}^{(1)}$ and $\overrightarrow{x}^{(2)}$ are linearly dependent.

Matrix and vector

Theorem (I)

Let A be an $n \times n$ matrix. Then the following are equivalent:

- *A is nonsingular/invertible.*
- *$det(A) \neq 0$.*
- *$\overrightarrow{x} = 0$ is the unique solution of $A\overrightarrow{x} = 0$.*
- *$A\overrightarrow{x} = \overrightarrow{b}$ has a unique solution.*
- *The columns (and rows) of A are linearly independent.*

Eigenvalue/eigenvector problems

Definition (Eigenvalue/eigenvector)

A value λ and a non-trivial vector $\overrightarrow{x}$ satisfying

$$A\overrightarrow{x} = \lambda\overrightarrow{x}$$

are called an eigenvalue and eigenvector of the matrix A respectively.

Eigenvalue/eigenvector problems

Remark

- *First,*

$$\begin{aligned} & A\vec{x} = \lambda\vec{x} \\ \Rightarrow \quad & A\vec{x} - \lambda\vec{x} = 0 \\ \Rightarrow \quad & (A - \lambda I)\vec{x} = 0 \end{aligned}$$

  where I is the identity matrix.

- *Since the eigenvector $\vec{x}$ needs to be non-trivial, then by Theorem (I), we obtain the characteristic equation of A:*

$$det(A - \lambda I) = 0.$$

Eigenvalue/eigenvector problems

- Example 7: Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}.$$

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- We compute the eigenvalues first:

$$0 = det(A - \lambda I)$$

Eigenvalue/eigenvector problems

- Example 7: Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}.$$

- We compute the eigenvalues first:

$$\begin{aligned} 0 &= det(A - \lambda I) \\ &= \begin{vmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{vmatrix} \end{aligned}$$

Eigenvalue/eigenvector problems

- Example 7: Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}.$$

- We compute the eigenvalues first:

$$\begin{aligned} 0 &= det(A - \lambda I) \\ &= \begin{vmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{vmatrix} \\ &= (3-\lambda)(-2-\lambda) + 4 \end{aligned}$$

Eigenvalue/eigenvector problems

- Example 7: Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}.$$

- We compute the eigenvalues first:

$$\begin{aligned} 0 &= det(A - \lambda I) \\ &= \begin{vmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{vmatrix} \\ &= (3-\lambda)(-2-\lambda) + 4 \\ &= -6 - 3\lambda + 2\lambda + \lambda^2 + 4 \\ &= \lambda^2 - \lambda - 2 \\ &= (\lambda - 2)(\lambda + 1). \end{aligned}$$

Eigenvalue/eigenvector problems

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- We compute the eigenvalues first:

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- Eigenvalues: $\lambda_1 = 2$ and $\lambda_2 = -1$.

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_1 = 2$:

$$(A - \lambda I)\vec{x} = 0$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_1 = 2$:

$$(A - \lambda I)\overrightarrow{x} = 0$$
$$\Rightarrow \quad (A - \lambda_1 I)\overrightarrow{\xi}^{(1)} = 0$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_1 = 2$:

$$
\begin{aligned}
& (A - \lambda I)\overrightarrow{x} = 0 \\
\Rightarrow \quad & (A - \lambda_1 I)\overrightarrow{\xi}^{(1)} = 0 \\
\Rightarrow \quad & \begin{pmatrix} 3-2 & -1 \\ 4 & -2-2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_1 = 2$:

$$\begin{aligned} & (A - \lambda I)\vec{x} = 0 \\ \Rightarrow \quad & (A - \lambda_1 I)\vec{\xi}^{(1)} = 0 \\ \Rightarrow \quad & \begin{pmatrix} 3-2 & -1 \\ 4 & -2-2 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \Rightarrow \quad & \begin{pmatrix} 1 & -1 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{aligned}$$

- Hence $\xi_1 - \xi_2 = 0$.

- Pick $\xi_2 = 1$, then $\xi_1 = 1$. Hence

$$\vec{\xi}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_2 = -1$:

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_2 = -1$:

$$(A - \lambda I)\overrightarrow{x} = 0$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_2 = -1$:

$$\begin{aligned} & (A - \lambda I)\vec{x} = 0 \\ \Rightarrow \quad & (A - \lambda_2 I)\vec{\xi}^{(2)} = 0 \end{aligned}$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_2 = -1$:

$$
\begin{aligned}
& (A - \lambda I)\vec{x} = 0 \\
\Rightarrow \quad & (A - \lambda_2 I)\vec{\xi}^{(2)} = 0 \\
\Rightarrow \quad & \begin{pmatrix} 3-(-1) & -1 \\ 4 & -2-(-1) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

Eigenvalue/eigenvector problems

- Now we compute the eigenvector corresponding to $\lambda_2 = -1$:

$$
\begin{aligned}
& (A - \lambda I)\vec{x} = 0 \\
\Rightarrow \quad & (A - \lambda_2 I)\vec{\xi}^{(2)} = 0 \\
\Rightarrow \quad & \begin{pmatrix} 3-(-1) & -1 \\ 4 & -2-(-1) \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\
\Rightarrow \quad & \begin{pmatrix} 4 & -1 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}
\end{aligned}
$$

Eigenvalue/eigenvector problems

- Hence $4\xi_1 - \xi_2 = 0$.

Eigenvalue/eigenvector problems

- Hence $4\xi_1 - \xi_2 = 0$.
- Pick $\xi_1 = 1$, then $\xi_2 = 4$. Hence

$$\vec{\xi}^{(2)} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

Special matrices

Definition (spectral radius)

$\rho(A) = \max_{1 \le i \le n} |\lambda_i|$, where $\{\lambda_i\}_{i=1}^n$ is the set of eigenvalues of A, is called the spectral radius of A.

Definition (symmetric matrix)

If $A^T = A$, then A is said to be symmetric.

Definition (Hermitian matrix)

If $A^H = A$, then A is said to be Hermitian.

Definition (positive definite matrix)

If $\overrightarrow{x}^T A \overrightarrow{x} > 0$ for any $\overrightarrow{x} \neq 0$, then A is said to be positive definite.

Special matrices

Theorem (II)

If A is symmetric positive definite or Hermitian positive definite, then its eigenvalues are real and positive.

Theorem (III)

If system $B\overrightarrow{x} = \overrightarrow{d}$ is obtained from system $A\overrightarrow{x} = \overrightarrow{b}$ by a finite sequence of elementary row operations, then the two systems have the same solutions.

Special matrices

Definition (Vandermonde matrix)

$$A = (a_{i-1}^{j-1})_{i=1,j=1}^{n+1,n+1} = \begin{pmatrix} 1 & a_0 & a_0^2 & a_0^3 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & a_1^3 & \cdots & a_1^n \\ & \vdots & & \ddots & & \\ 1 & a_n & a_n^2 & a_n^3 & \cdots & a_n^n \end{pmatrix}.$$

Remark

The determinant of the Vandermonde matrix is

$$det(A) = \prod_{k=1}^{n} \prod_{j=0}^{k-1} (a_k - a_j) \quad .$$